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# Batch 15

# Algorithm

1. Start.
2. Read two positive integers `a` and `b`.
3. Call the recursive function `gcd(a, b)`.
4. In `gcd(a, b)`:
   - If `b == 0`, return `a`.
   - Otherwise, return `gcd(b, a % b)`.
5. Display the GCD.
6. Stop.

# Test Case Table

| Test Case | a | b | Expected GCD |
|---|---|---|---|
| 1 | 48 | 18 | 6 |
| 2 | 20 | 15 | 5 |
| 3 | 17 | 13 | 1 |
| 4 | 100 | 25 | 25 |
| 5 | 7 | 7 | 7 |

# C Program

```
#include <stdio.h>

int gcd(int a, int b)
{
    if (b == 0)
        return a;
```

```
    return gcd(b, a % b);
}

int main()
{
    int a, b, result;

    printf("Enter two positive integers: ");
    scanf("%d %d", &a, &b);

    result = gcd(a, b);

    printf("GCD = %d\n", result);

    return 0;
}
```

# Execution

